

ASSIGNMENT 2

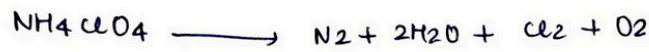
Reactor Design Analysis

Influence of Reactor Selection and Configuration on Performance

Field	Details
Group	Group 8
Reaction System	Decomposition of Ammonium Perchlorate (NH_4ClO_4)
Application	Rocket fuel propellant decomposition
Rate Law	$r = k \cdot [\text{NH}_4\text{ClO}_4]^{1/2}$ - Half-order reaction
Rate constant	$k = 2.9 \times 10^{-3} \text{ mol}^{1/2} \text{ L}^{-1/2} \text{ s}^{-1}$ at 600K
Activation energy	$E_a = 120 \text{ kJ/mol}$

1. Introduction to Problem Statement:

Ammonium perchlorate (NH_4ClO_4) is the primary oxidizer in solid fuel rocket propellants. Its thermal decomposition is one of the most important rxn in aerospace engg., governing the burn rate, efficiency & stability of rocket boosters, so industry's significant research interest is understanding the kinetics & designing the appropriate reactor



~~The reaction was NH4ClO4~~
The decomposition follows a half-order rate law w.r.t. NH_4ClO_4 , which is characteristic of heterogeneous or surface-controlled processes. For this analysis, the reaction is modelling as a pseudo-homogeneous liquid-phase process with the given kinetic parameters, as is standard practice in reactor design course work.

#2. Assumptions & Base Case Parameters:

No.	Assumption / Parameter	Value / Justification
1	Initial concentration (C_{a0})	1.0 mol/L of NH_4ClO_4
2	Volumetric feed flow rate (v_0)	1.0 L/s
3	Molar feed rate (F_{a0})	$F_{a0} = C_{a0} \times v_0 = 1.0 \text{ mol/s}$
4	Target final conversion (X)	80% ($X = 0.80$) unless stated otherwise
5	Operating temperature	600 K (isothermal) unless studying temperature effects
6	Reactor <u>ideality</u>	Perfect mixing for CSTR; perfect plug flow for PFR
7	Steady-state operation	All reactors operate at steady state
8	Single reactant	NH_4ClO_4 is the rate-limiting species (A); others are in excess or not tracked
9	Negligible pressure <u>drop</u>	Liquid-phase — pressure effects on density are negligible

GOVERNING EQUATIONS

• Rate Law

$$-r_A = k \cdot C_A^{1/2} = k [C_{A0}(1-X)]^{1/2}$$

• Arrhenius Eqn:

$$k(T) = A \exp[-E_a/RT]$$

$$k(T) = k(T_{ref}) \cdot \exp\left[\frac{E_a}{R} \cdot \left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right]$$

$$E_a = 120 \text{ KJ}$$

$$\frac{E_a}{R} = \frac{120000}{8.314} = 14,430.6 \text{ K}, \quad T_{ref} = 600 \text{ K}, \quad k_{ref} = 2.9 \times 10^{-3}$$

• CSTR Design Eqn (mole balance):

$$V_{CSTR} = \frac{F_{A0} \cdot X}{(-r_A)_{exit}} = \frac{F_{A0} \cdot X}{[k C_{A0}^{1/2} (1-X)^{1/2}]} \quad (\text{for half order rxn})$$

• PFR Design Eqn (mole balance):

$$V_{PFR} = F_{A0} \int_{x_1}^{x_2} \frac{dx}{(-r_A)} = F_{A0} \int_{x_1}^{x_2} \frac{dx}{k C_{A0}^{1/2} (1-x)^{1/2}}$$

$$= \frac{2 F_{A0}}{k C_{A0}^{1/2}} \left(\frac{\sqrt{1-x_1}}{\sqrt{x_1}} - \frac{\sqrt{1-x_2}}{\sqrt{x_2}} \right)$$

• Key Design Parameter

$$T = 600 \text{ K}, \quad k = 2.9 \times 10^{-3}, \quad C_{A0} = 1.0 \text{ mol/L}, \quad F_{A0} = 1 \text{ mol/s}$$

$$F_{A0} / (k \cdot C_{A0}^{1/2}) = 1 / (2.9 \times 10^{-3} \times 1) = 344.83 \text{ L} \quad (= 344.83 \text{ s})$$

①. Effect of Temperature on Rate constant:

$$A = k_{ref} \cdot \exp \left(\frac{E_a}{R(T_{ref})} \right) = (2.9 \times 10^{-3}) \cdot \exp \left(\frac{120000}{8.314 \times 600} \right)$$

$$= 8.09 \times 10^7 \text{ mol}^{1/2} \text{ L}^{-1/2} \text{ s}^{-1}$$

$$k = A \exp \left(-\frac{E_a}{RT} \right)$$

$$\frac{k_1}{k_2} = \exp \left[\frac{-E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \right]$$

Temperature	Temperature (°C)	Rate Constant k	k(T) / k(600 K)	% Change vs 600 K
500 K	226.9 °C	2.360×10^{-5}	8.138e-3	-99.2%
550 K	276.9 °C	3.256×10^{-4}	1.123e-1	-88.8%
575 K	301.9 °C	1.019×10^{-3}	3.514e-1	-64.9%
600 K	326.9 °C	2.900×10^{-3}	1.000e+0	0.0%
625 K	351.9 °C	7.591×10^{-3}	2.618e+0	161.8%
650 K	376.9 °C	1.845×10^{-2}	6.363e+0	536.3%
700 K	426.9 °C	9.013×10^{-2}	3.108e+1	3007.9%
750 K	476.9 °C	3.563×10^{-1}	1.229e+2	12187.4%
800 K	526.9 °C	1.186×10^0	4.091e+2	40809.7%

Key Observation:

→ The rate constant increase by approx 3x when temp rises from 600K to 700K (only 100K increase), and by over 400x from 600K to 800K. This is direct consequence of the high activation energy. Such strong sensitivity means that temp control is most powerful lever for adjusting reaction rates in system.

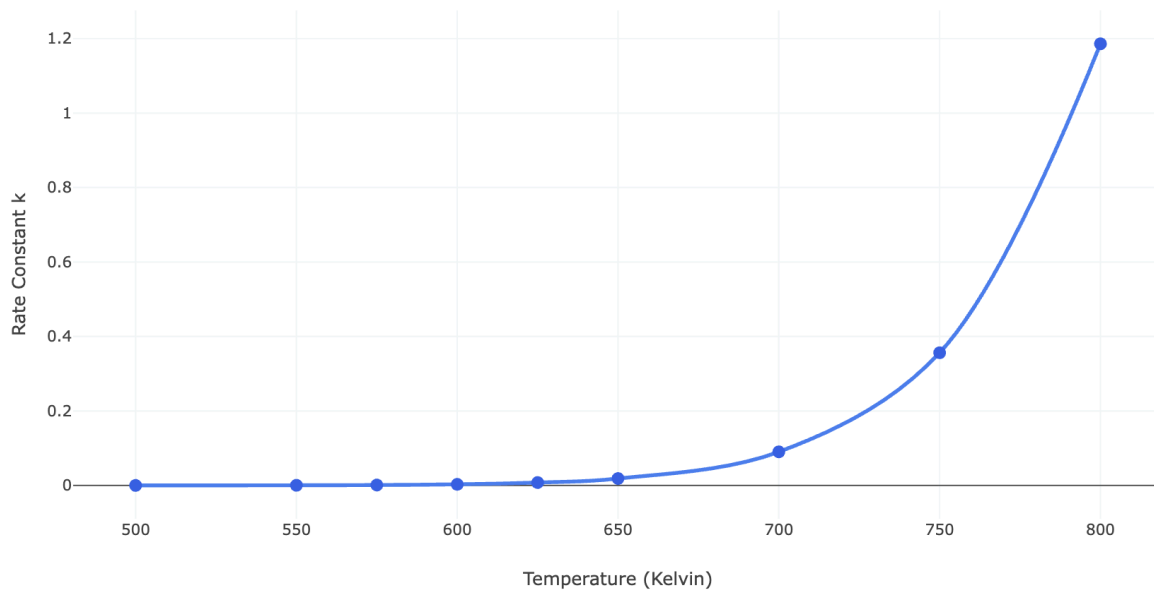
The Arrhenius plot ($\ln k$ vs $1/T$) yielded straight line with slope $= -\frac{E_a}{R} = -14,430 \text{ K}$. The linearity of this plot is hallmark of elementary Arrhenius behaviour & validates kinetic model provided.

Rate Constant Analysis

Visualization of the relationship between Temperature (K) and Rate Constant (k).

Linear Scale

Logarithmic Scale

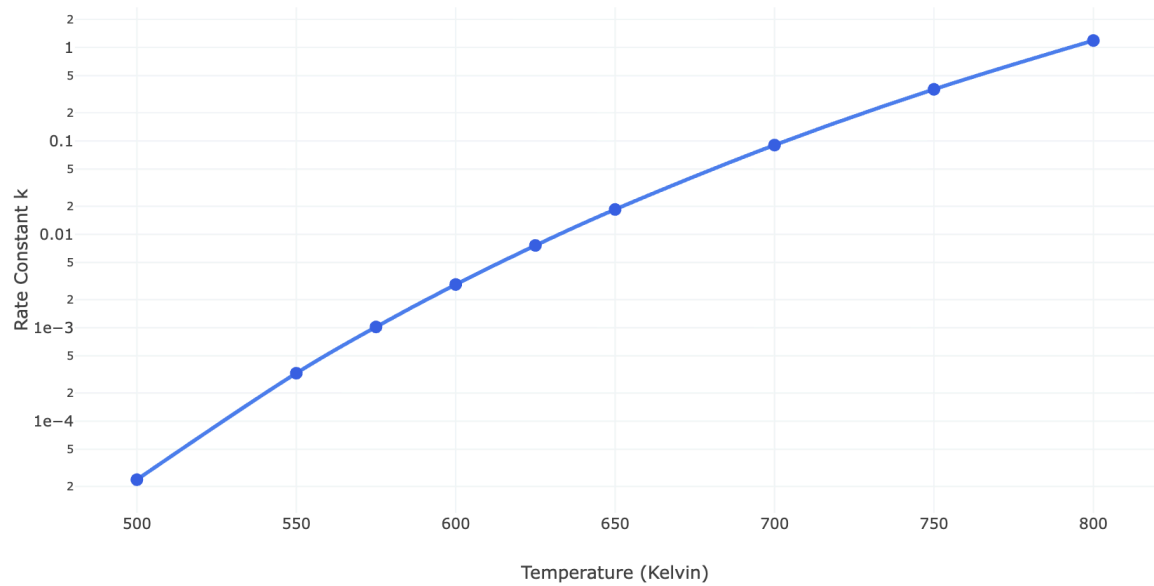


Rate Constant Analysis

Visualization of the relationship between Temperature (K) and Rate Constant (k).

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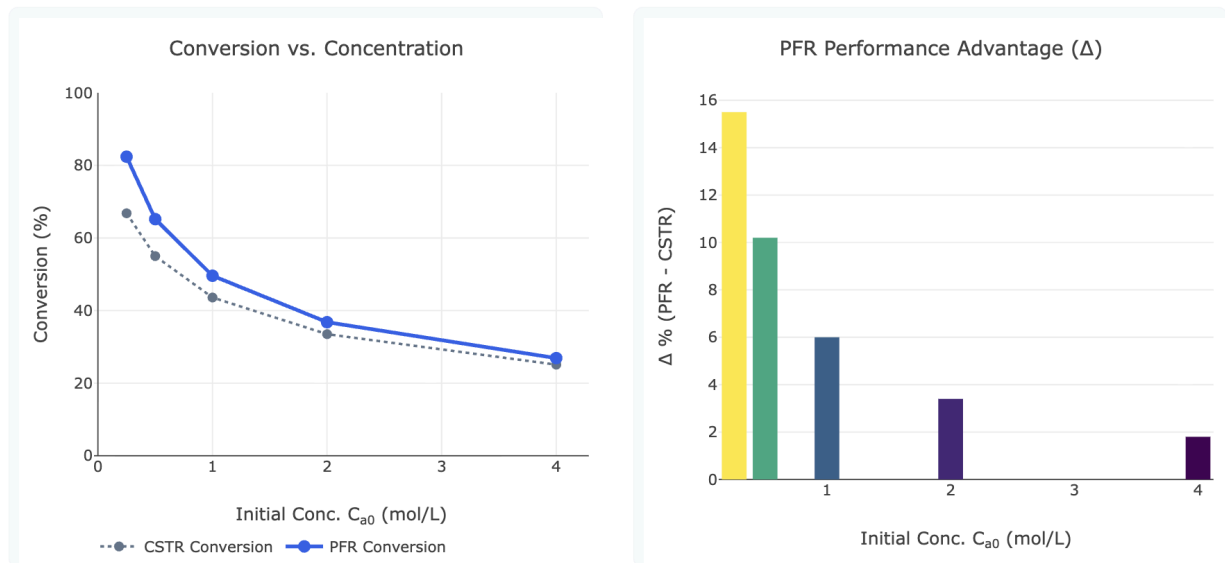


5) EFFECT OF CONC^N ON CONVERSION.

Initial Conc. C_{a_0}	$k\tau/C_{a_0}^{1/2}$ (CSTR param.)	CSTR Conversion	PFR Conversion	Δ (PFR - CSTR)
0.25 mol/L	1.160	66.8%	82.4%	15.5%
0.5 mol/L	0.820	55.0%	65.2%	10.2%
1 mol/L	0.580	43.6%	49.6%	6.0%
2 mol/L	0.410	33.5%	36.8%	3.4%
4 mol/L	0.290	25.1%	26.9%	1.8%

Reactor Performance Comparison

Analysis of CSTR vs. PFR Conversion across varying Initial Concentrations (C_{a0}).



★ Conclusion

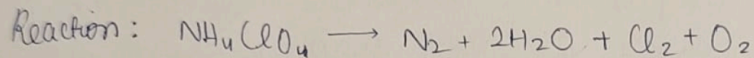
For $1/2$ order rxn, conversion dec. as C_{a0} inc. (at fixed τ). This is because rate $r = k \cdot C_a^{1/2}$ grows more slowly than linearly with concⁿ. If C_{a0} doubles, rate only inc. by a factor of $\sqrt{2}$, so the reactor must process a disproportionately larger molar load for same volume.

PFR consistently achieves higher conversion than CSTR for all concⁿs as PFR maintains higher driving force, while CSTR operates entirely at low exit concⁿ.

Summary: effect of Pressure on Gas-Phase Ammonium Perchlorate Decomposition

This addendum analyzes how pressure and volumetric expansion affect the decomposition of Ammonium Perchlorate (AP) when modeled as gas phase process.

1) STOICHIOMETRY AND EXPANSION (E)



Expansion parameter (E) $\rightarrow$ every 1 mole of reactant produces 5 moles of gas, resulting in $E = +4$.

Impact: At complete conversion ($X=1$), the gas expands to five times its original volume at constant pressure.

This rapid expansion dilutes the remaining reactant and reduces the reaction rate.

2) CONCENTRATION AND PRESSURE EFFECTS:

The concentration of reactant A (C_A) is governed by the pressure ratio ($\Phi = P/P_0$) and conversion (X)

$$C_A = C_{A0} \Phi \frac{1-X}{1+4X}$$

- Kinetic benefit: higher pressure (Φ) directly increases concentration.
- Expansion Penalty: The $(1+4X)$ term shows that gas expansion dilutes the reactant much more aggressively than in liquid-phase reactions.

6. comparison of reactor configuration:

$$C_{A0} = 1 \text{ mol/L}, V_0 = 1 \text{ L/s}, T = 600 \text{ K}, \text{ target } X = 0.80$$

• 1. single CSTR

$$V_{\text{CSTR}} = \frac{(1.0 \times 80.80)}{(2.9 \times 10^{-3} \times 1^{1/2} (0.2)^{1/2})} = 616.8 \text{ L}$$

• single PFR:

$$V_{\text{PFR}} = \left(\frac{F_{A0} (2)}{k_0 \cdot C_{A0}^{1/2}} \right) (1 - \sqrt{1 - 0.80})$$

$$= (344.83) (2) (1 - 0.4472)$$

$$= (344.83 \times 1.056)$$

• ~~CSTR~~ ~~for~~ = 381.2 L

• CSTR followed by PFR (CSTR $\rightarrow$ PFR), $X_1 = 0.50$)

$$V_{\text{CSTR}} = 344.83 \times \frac{0.50}{(0.50)^{1/2}} = \frac{172.4}{0.7071} = 243.8 \text{ L}$$

$$V_{\text{PFR}} = 344.83 \times 2 \times [\sqrt{0.50} - \sqrt{0.20}] = 689.66 \times 0.2599 = 179.2 \text{ L}$$

$$V_{\text{Total}} = 243.8 + 179.2 = 423.0 \text{ L}$$

• PFR followed by CSTR (PFR $\rightarrow$ CSTR, $X_1 = 0.50$)

$$V_{\text{PFR}} = 344.83 \times 2 \times (1 - \sqrt{0.50}) = 689.66 \times 0.2929 = 202.0 \text{ L}$$

$$V_{\text{CSTR}} = (344.83) \frac{(0.8 - 0.50)}{(0.2)^{1/2}} = \frac{103.4}{0.4472} = 231.3 \text{ L}$$

$$V_{\text{Total}} = 202.0 + 231.3 = 433.3 \text{ L}$$

• Big CSTR followed by small CSTR ($X_1 = 0.65, X_2 = 0.80$)

$$V_1(\text{big}) (0 \rightarrow 0.65) = \frac{344.83 \times 0.65}{(1 - 0.65)^{1/2}} = \frac{224.1}{0.5916} = 378.9 \text{ L}$$

$$V_2(\text{small}) = \frac{344.83 (0.80 - 0.65)}{(1 - 0.80)^{1/2}} = \frac{51.72}{0.4472} = 115.7 \text{ L}$$

~~(0.80 $\rightarrow$ 0.65)~~
(0.65 $\rightarrow$ 0.80)

$$V_{\text{Total}} = 378.9 + 115.7 = 494.6 \text{ L}$$

◦ small CSTR followed by Big CSTR ($X_1 = 0.15$, $X_2 = 0.80$)

$$V_1 (\text{small}) = 344.83 \times \frac{0.15}{(1-0.15)^{1/2}} = \frac{51.72}{0.9220} = 56.11$$

$$V_2 (\text{big}) = 344.83 \times \frac{(0.8 - 0.15)}{(1-0.8)^{1/2}} = \frac{224.1}{0.4472}$$

$$= 501.2 \text{ L}$$

$$V_{\text{Total}} = 56.1 + 501.2 = 557.3 \text{ L}$$

◦ Two equal volume CSTRs in series

$$\frac{X_1}{(1-X_1)^{1/2}} = \frac{X_2 - X_1}{(1-X_2)^{1/2}}$$

$$X_1 = 0.80 \Rightarrow \boxed{X_1 = 0.4916}$$

$$V_1 = V_2 = 237.8 \text{ L}$$

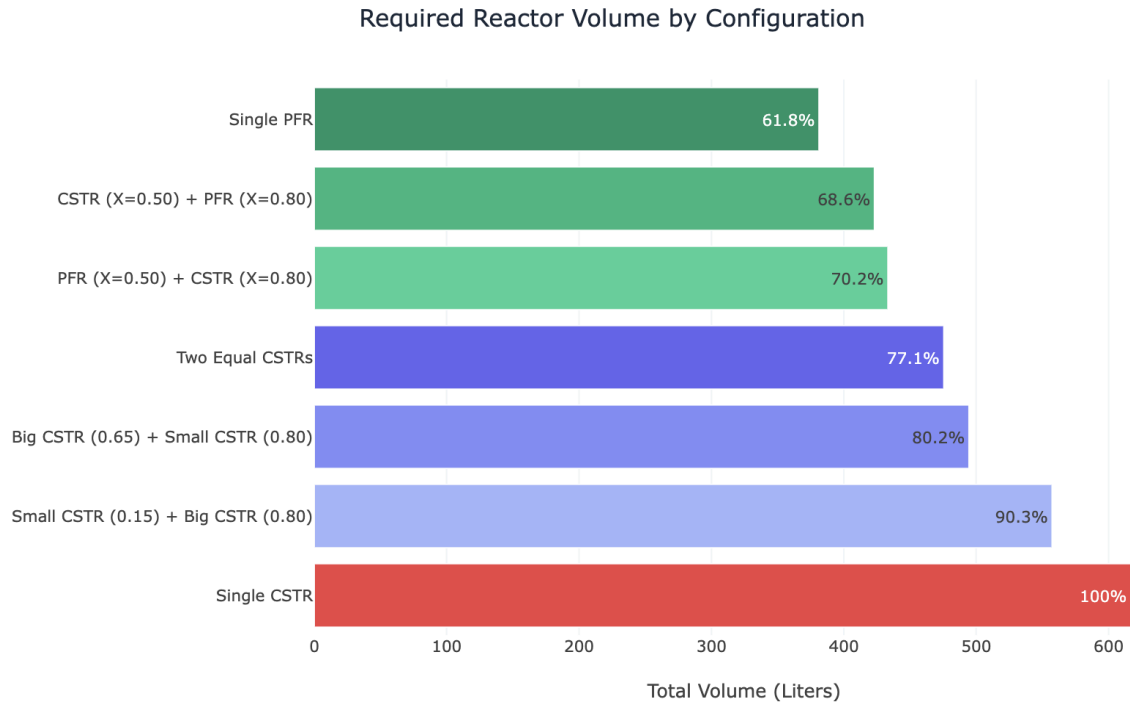
$$V_{\text{Total}} = 2(237.8) = 475.5 \text{ L}$$

6.8 Summary: All Configurations Ranked by Total Volume

Rank	Configuration	Stage Volumes	Total Volume	% of Single CSTR
1	Single PFR	381.2 L	381.2 L	61.8%
2	CSTR ($X=0.50$) + PFR ($X=0.80$)	243.8 L + 179.2 L	423.1 L	68.6%
3	PFR ($X=0.50$) + CSTR ($X=0.80$)	202.0 L + 231.3 L	433.3 L	70.2%
4	Two Equal CSTRs ($X_1 \approx 0.492$, $X_2 = 0.80$)	237.8 L + 237.8 L	475.5 L	77.1%
5	Big CSTR ($X=0.65$) + Small CSTR ($X=0.80$)	378.9 L + 115.7 L	494.5 L	80.2%
6	Small CSTR ($X=0.15$) + Big CSTR ($X=0.80$)	56.1 L + 501.2 L	557.3 L	90.3%
7	Single CSTR	616.8 L	616.8 L	100.0%

Reactor Network Optimization

Comparison of total volume requirements for reaching 80% conversion ($X=0.80$).



★ Conclusion

The PFR is the most volume efficient reactor for this $1/2$ order positive rate kx^n . Hybrid & staged configurations occupy the middle ground, & their ranking follows from how much of the kx^n occurs in the efficient PFR mode. In all cases, staging is better than a single CSTR.

7) INCREASING TEMP. V/S INC. VOLUME

To improve conversion, we might either try to inc. temp. or reactor volume as there'll be a tradeoff in terms of cost & energy too.

A) Increasing Temp. at fixed volume

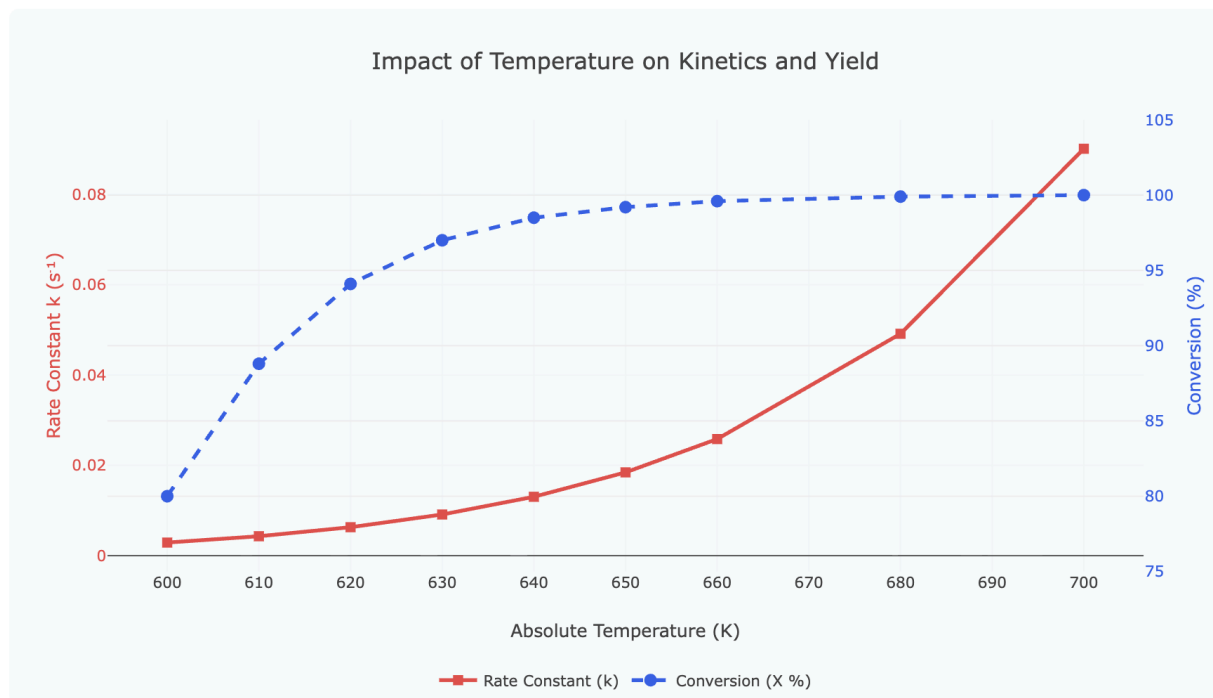
Base case $V = 616.8 \text{ L}$, $T = 600 \text{ K}$, $x = 0.80$.

Temp is raised in steps while $V = \text{const} = 616.8 \text{ L}$

Temperature	ΔT from Base	Rate Constant k	Conversion X
600 K	0 K	2.900×10^{-3}	80.0%
610 K	10 K	4.302×10^{-3}	88.8%
620 K	20 K	6.301×10^{-3}	94.1%
630 K	30 K	9.118×10^{-3}	97.0%
640 K	40 K	1.304×10^{-2}	98.5%
650 K	50 K	1.845×10^{-2}	99.2%
660 K	60 K	2.583×10^{-2}	99.6%
680 K	80 K	4.915×10^{-2}	99.9%
700 K	100 K	9.013×10^{-2}	100.0%

Temperature Sensitivity Analysis

Effect of ΔT from base (600 K) on Rate Constant (k) and Conversion (X).

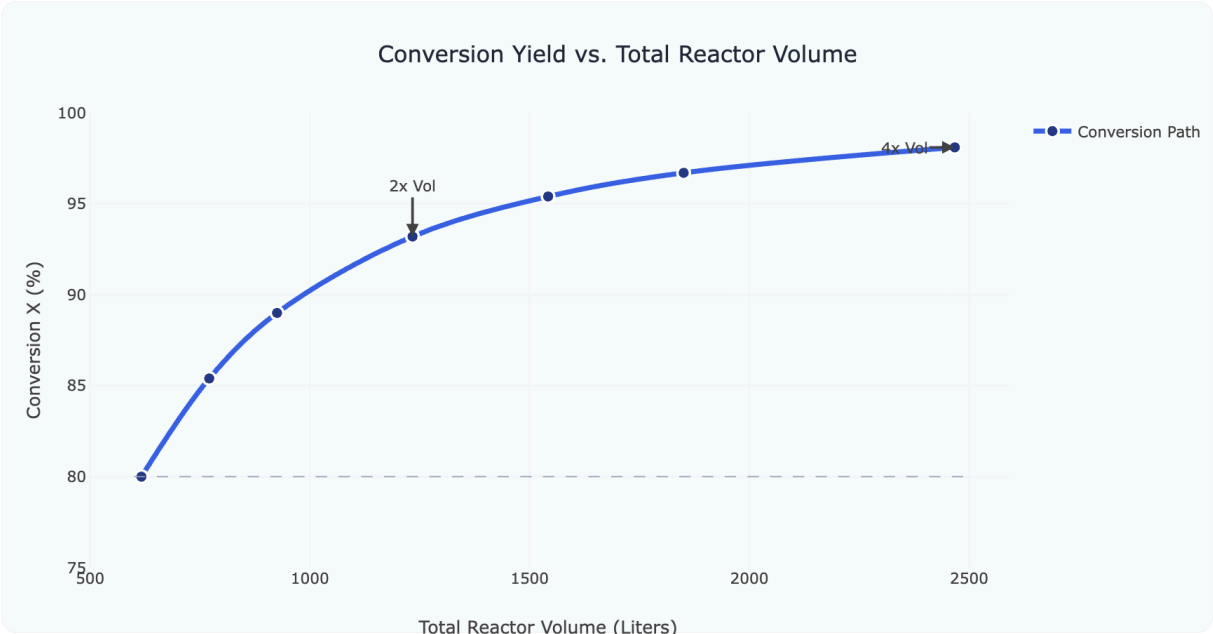


B) Increasing Volume (Fixed Temp = 600K)
 Temp is held fixed at 600K & CSTR volume is increased as a multiple of base 616.8L

Volume Multiplier	Total Volume	Extra Volume Added	Conversion X
1.00x	616.8 L	+0.0 L	80.0%
1.25x	771.1 L	+154.2 L	85.4%
1.50x	925.3 L	+308.4 L	89.0%
2.00x	1233.7 L	+616.8 L	93.2%
2.50x	1542.1 L	+925.3 L	95.4%
3.00x	1850.5 L	+1233.7 L	96.7%
4.00x	2467.4 L	+1850.5 L	98.1%

Reactor Sizing & Conversion Analysis

Evaluating the impact of volume expansion on process yield (Conversion X).



BASE CASE (1.0X)

80.0% X

616.8 L

DOUBLED VOLUME (2.0X)

93.2% X

$\Delta X = +13.2\%$

MAX EFFORT (4.0X)

98.1% X

4x volume for +18.1% X

7.3 Comparative Conclusion

Strategy	Change Required	Achieves X = 90%	Achieves X = 95%	Recommendation
Raise Temperature	+~10–20 K	Yes (+10 K)	Yes (+20 K)	✓ PREFERRED
Increase Volume	+~50% volume	+50% volume	+130% volume	Less efficient

* Conclusion

On increasing temp, (from 600K to just 620K), conversion is boosted from 80% to 94.1%. Further inc. to 650K gives 99.2% conversion, i.e., nearly complete rxn.

However, tripling reactor volume (from 617L to 1850L) only inc. conversion from 80% to 96.7%. Going beyond 2.5x yields sharply diminishing returns due to $\frac{1}{2}$ order kinetics.

$\therefore$ Inc. T is dramatically more effective than inc. reactor volume.

8) IMPACT OF FLOW RATE ON CONVERSION

For a fixed reactor volume, the volumetric flow rate v_0 determines residence time $T = V/v_0$, which in turn controls conversion. Higher flow rates mean shorter residence times & lower conversion. Both CSTR ($V = 66.8 \text{ L}$) & PFR ($V = 381.2 \text{ L}$) as flow rate is varied from 0.3 to 4 L/s.

→ For CSTR, operating eqn at varying v_0 is:

$$\frac{x}{(1-x)^{1/2}} = \frac{K \cdot V_{\text{CSTR}}}{v_0 \cdot (C_{A0})^{1/2}} = \frac{K \cdot T}{(C_{A0})^{1/2}}$$

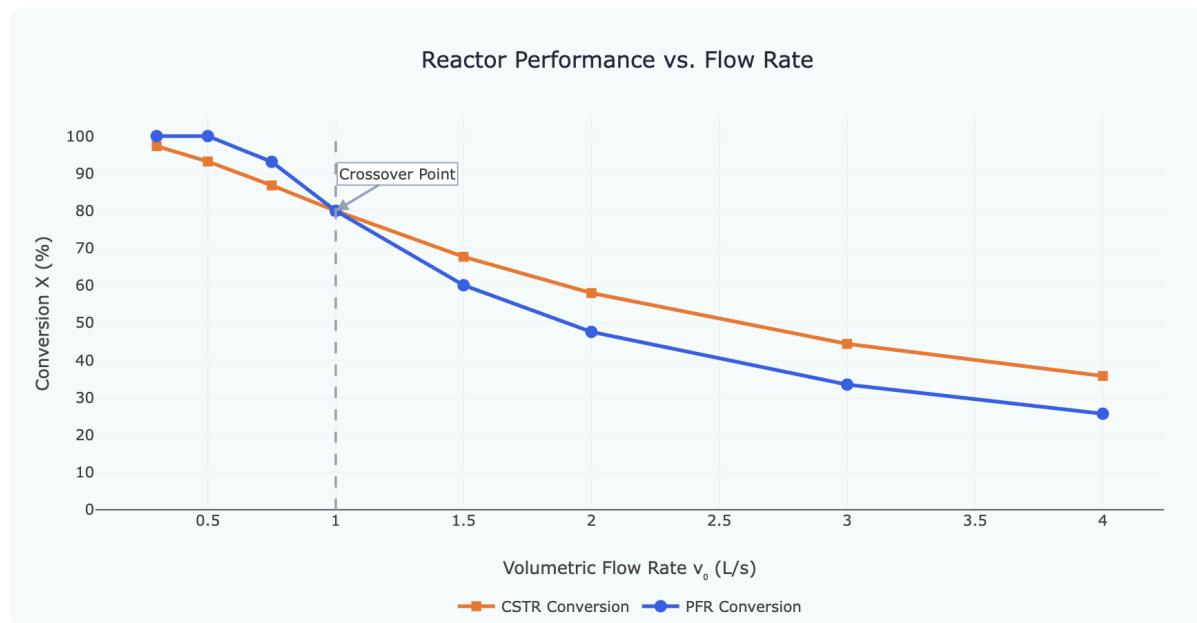
→ For PFR, eqn is :-

$$x = 1 - \left[\frac{1 - Kx \cdot V_{\text{PFR}}}{2x v_0 x (C_{A0})^{1/2}} \right]^2$$

Flow Rate v_0 (L/s)	τ_{CSTR} (s)	τ_{PFR} (s)	CSTR Conversion	PFR Conversion
0.30 L/s	2056.2 s	1270.8 s	97.3%	≈100.0%
0.50 L/s	1233.7 s	762.5 s	93.2%	≈100.0%
0.75 L/s	822.5 s	508.3 s	86.8%	93.1%
1.00 L/s	616.8 s	381.2 s	80.0%	80.0%
1.50 L/s	411.2 s	254.2 s	67.7%	60.1%
2.00 L/s	308.4 s	190.6 s	58.0%	47.6%
3.00 L/s	205.6 s	127.1 s	44.4%	33.5%
4.00 L/s	154.2 s	95.3 s	35.8%	25.7%

Flow Rate & Residence Time Analysis

Comparing CSTR and PFR conversion performance across varying flow rates (v_0).



KEY OBSERVATION: THE CROSSOVER POINT

At flow rates below **1.0 L/s**, the PFR is significantly more efficient. However, at flow rates above **1.0 L/s**, the CSTR provides higher conversion. This inversion at $v_0 = 1.0$ suggests that the reactor performance is highly sensitive to residence time distribution for this specific reaction.

Conclusion

At design flow rate of $v_0 = 1 \text{ L/s}$, both reactors achieve the target $x = 80\%$. As flow rate dec. below this value, conversion rises in both reactors. — the PFR more sharply, reaching near-complete conversion of $v_0 = 0.5 \text{ L/s}$. At high flow rates, both reactors lose conversion rapidly. However, PFR always outperforms the CSTR at any given flow rate.

9) Economic Analysis

An economic analysis compares the total cost of ownership for different reactor configurations. Cost is composed of capital cost (proportional to reactor volume, estimated by the six-tenths rule) and operating cost (driven primarily by energy to maintain temperature).

9.1) Capital Cost Model

The six-tenths rule estimates purchased equipment cost (PEC) for reactors:

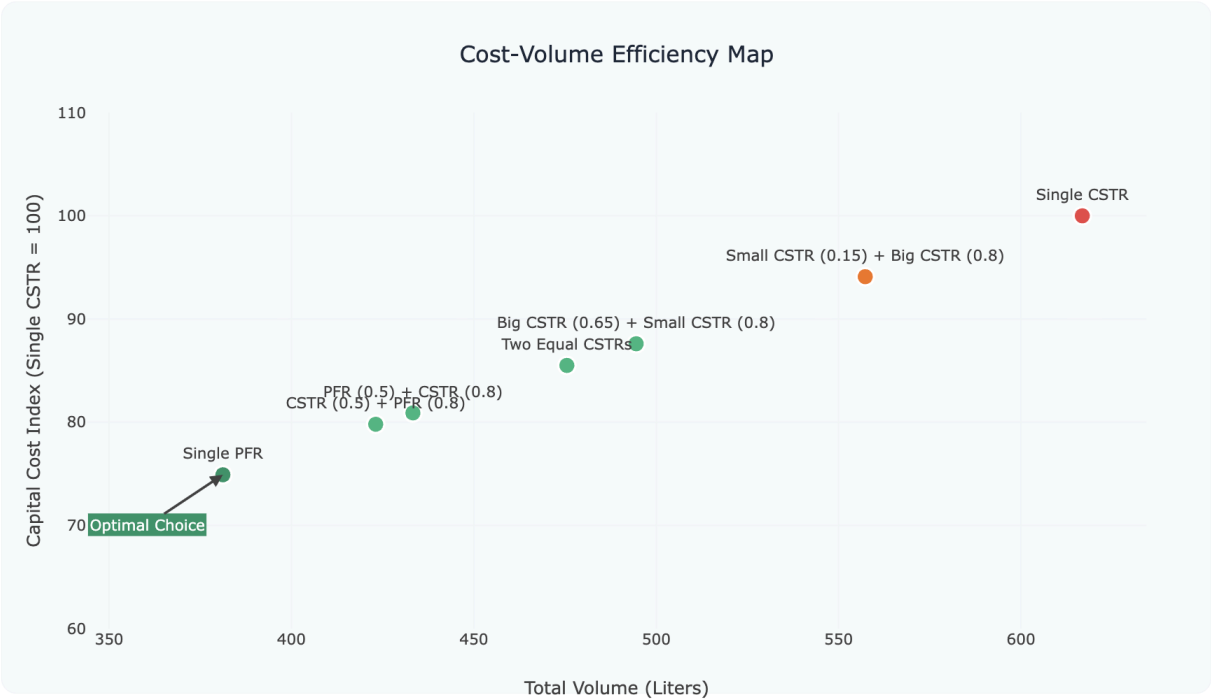
$$PEC \propto V^{0.6}$$

All capital costs are normalized to the single CSTR (616.8 L) = 100 index units.

Configuration	Total Volume	Capital Cost Index (CSTR = 100)	Capital Tier
Single CSTR	616.8 L	100.0	Moderate
Single PFR	381.2 L	74.9	Low (✓)
CSTR (X=0.50) + PFR (X=0.80)	423.1 L	79.8	Low (✓)
PFR (X=0.50) + CSTR (X=0.80)	433.3 L	80.9	Low (✓)
Big CSTR (X=0.65) + Small CSTR (X=0.80)	494.5 L	87.6	Low (✓)
Small CSTR (X=0.15) + Big CSTR (X=0.80)	557.3 L	94.1	Moderate
Two Equal CSTRs (X1≈0.492, X2=0.80)	475.5 L	85.5	Low (✓)

Capital Cost & Volume Analysis

Economic comparison of reactor configurations relative to a single CSTR benchmark (Index = 100).



ECONOMIC TAKEAWAY

The **Single PFR** is the clear economic winner, requiring the least volume and the lowest capital cost index (74.9). Note that configurations in the **"Low" Capital Tier** (green markers) all cluster towards the lower-left, representing higher efficiency designs.

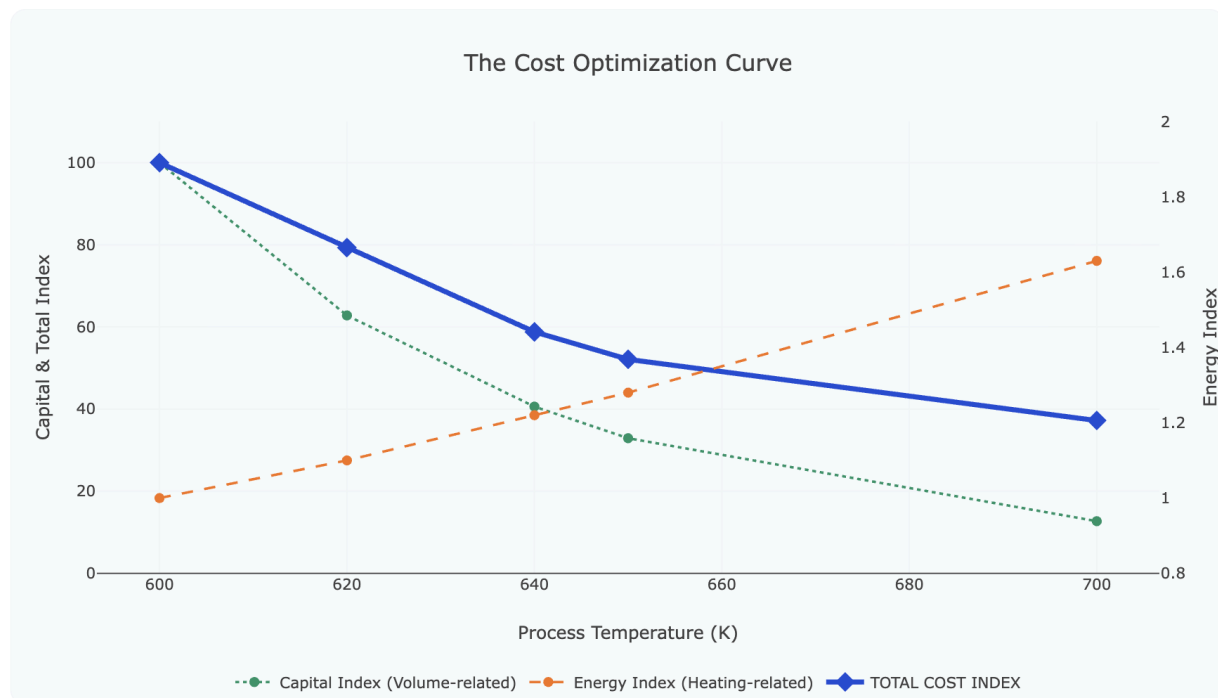
9.2) Operating Cost and Energy

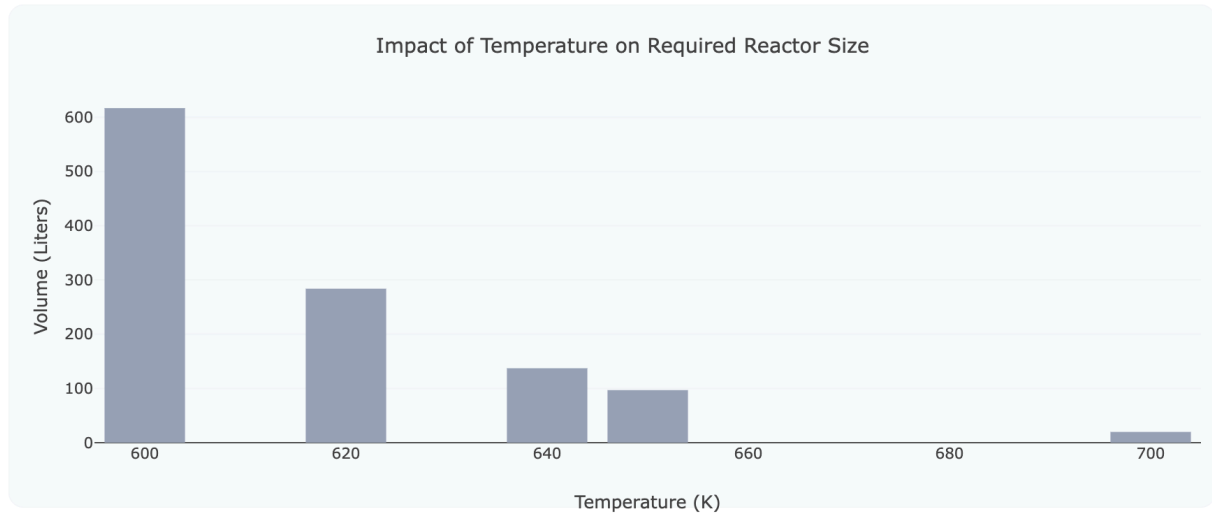
Operating at higher temperature reduces the required reactor volume (and hence capital cost) but increases energy costs. For the NH_4ClO_4 decomposition, a rough energy cost index (relative to 600K=1.0) is estimated below, assuming 5% additional energy cost per 10K rise:

Temperature	k (rate const.)	Volume for X=80%	Capital Index	Energy Index	Total Cost Index
600 K	2.90×10^{-3}	616.8 L	100.0	1.00	100.0
620 K	6.30×10^{-3}	283.9 L	62.8	1.10	79.3
640 K	1.30×10^{-2}	137.2 L	40.6	1.22	58.8
650 K	1.85×10^{-2}	96.9 L	32.9	1.28	52.1
700 K	8.99×10^{-2}	19.8 L	12.7	1.63	37.2

Total Cost Optimization

Balancing Reactor Volume (Capital) against Heating Requirements (Energy) to find the optimal operating temperature.





Economic conclusion: Moderately increasing temperature (e.g., from 600 K to 640-650 K) dramatically reduces capital cost (smaller reactor needed) while the incremental energy cost increase is relatively small. Beyond ~ 680 K, energy costs begin to dominate and the benefit of further temperature increase diminishes. The economically optimal operating temperature for this system is approximately 630-650 K for a single CSTR setup.

10. Conclusion:

This report has conducted a comprehensive reactor design analysis for the decomposition of ammonium perchlorate, a half order ($n = 0.5$) reaction with $E_a = 120 \text{ kJ/mol}$. at $T_{ref} = 600 \text{ K}$. The key findings are summarised below:

Analysis Area	Key Conclusion
Temperature Sensitivity	With $E_a = 120 \text{ kJ/mol}$, the rate constant increases $31\times$ over just 100 K ($600 \rightarrow 700 \text{ K}$). The reaction is highly temperature-sensitive.
Concentration Effect	Higher C_{a0} DECREASES conversion at fixed τ for this half-order reaction. Lower concentrations or longer residence times are preferred at high feed concentrations.
Best Single Reactor	PFR (381.2 L) requires 38.2% less volume than CSTR (616.8 L) for $X = 80\%$.
Best Hybrid Config	CSTR $\rightarrow$ PFR series (total 423.1 L) outperforms PFR $\rightarrow$ CSTR (433.3 L), since the PFR at the end handles high-conversion (low-rate) region efficiently.
CSTR Staging	Two equal CSTRs in series (475.5 L) save $\sim 22.9\%$ volume vs single CSTR. More staging approaches PFR performance.
Temperature vs Volume	Raising T by $\sim 20 \text{ K}$ is equivalent (in conversion gain) to more than doubling reactor volume. Temperature is the more effective lever.
Flow Rate	Conversion drops steeply with increasing flow rate. The PFR is always more resilient to throughput increases than the CSTR.
Economic Optimum	Moderately elevated temperature ($630\text{--}650 \text{ K}$) with a PFR or CSTR $\rightarrow$ PFR configuration offers the best balance of capital and operating cost.

Overall, the single PFR with a moderately elevated temperature ($630\text{--}650 \text{ K}$) represents the most efficient design for the decomposition of ammonium perchlorate to the target 80% conversion, where continuous-flow operation is impractical for this energetic material, a CSTR $\rightarrow$ PFR series at $620\text{--}630 \text{ K}$ represents a sound and economically justified alternative.

11. Comprehensive Summary of All Calculated Values

11.1 Arrhenius Calculation Details

T (K)	Ea/R (K)	(1/Tref-1/T) (K ⁻¹)	Exponent	k (mol ^{0.5} L ^{-0.5} s ⁻¹)
500	14,430.6	-3.3333e-4	-4.811	2.3601e-5
550	14,430.6	-1.5152e-4	-2.187	3.2557e-4
575	14,430.6	-7.2464e-5	-1.046	1.0190e-3
600	14,430.6	0.0000e+0	0.000	2.9000e-3
625	14,430.6	6.6667e-5	0.962	7.5908e-3
650	14,430.6	1.2821e-4	1.850	1.8452e-2
700	14,430.6	2.3810e-4	3.437	9.0130e-2
750	14,430.6	3.3333e-4	4.811	3.5634e-1
800	14,430.6	4.1667e-4	6.014	1.1864e+0

11.2 Reactor Volume Summary (X = 0.80, k = k_600)

Configuration	V ₁ (L)	V ₂ (L)	V _{total} (L)	Savings vs Single CSTR
Single CSTR	616.85	—	616.85	0.00%
Single PFR	381.23	—	381.23	38.20%
CSTR (X=0.50) + PFR (X=0.80)	243.83	179.24	423.07	31.41%
PFR (X=0.50) + CSTR (X=0.80)	202.00	231.32	433.31	29.75%
Big CSTR (X=0.65) + Small CSTR (X=0.80)	378.86	115.66	494.52	19.83%
Small CSTR (X=0.15) + Big CSTR (X=0.80)	56.10	501.19	557.29	9.65%
Two Equal CSTRs (X1≈0.492, X2=0.80)	237.77	237.77	475.54	22.91%

11.3 Concentration Effect Summary (τ = 200 s, T = 600 K)

C _{a0} (mol/L)	kτ/C _{a0} ^{0.5}	CSTR Conv.	PFR Conv.	PFR Advantage
0.25	1.160	66.82%	82.36%	+15.54 pp
0.5	0.820	55.01%	65.20%	+10.19 pp
1	0.580	43.57%	49.59%	+6.02 pp
2	0.410	33.46%	36.81%	+3.35 pp
4	0.290	25.10%	26.90%	+1.80 pp

11.4 Flow Rate Effect Summary

v ₀ (L/s)	τ _{CSTR} (s)	τ _{PFR} (s)	CSTR Conv.	PFR Conv.
0.30	2056.15	1270.77	97.34%	~100.00
0.50	1233.69	762.46	93.21%	~100.00
0.75	822.46	508.31	86.77%	93.09%
1.00	616.85	381.23	80.00%	80.00%
1.50	411.23	254.15	67.74%	60.12%
2.00	308.42	190.62	57.98%	47.64%
3.00	205.62	127.08	44.44%	33.46%
4.00	154.21	95.31	35.83%	25.73%